



Overview of Modern Component Failures in Rail Transit Environment

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B O S T O N



Executive Summary



Modern rail systems are complex and interconnected. Wheel-rail forces, vehicle dynamics, materials, maintenance, and operating conditions interact to produce failures that are not always apparent from the failed component alone.

The goal of failure analysis isn't just to explain the past, but to also prevent the next failure.

Understanding the root causes and contributing factors can help prevent costly service disruptions, improve reliability, and reduce the potential for safety incidents.



Case Study 1: Failure Analysis of Passenger Wheel Tires



Background:

Four derailments had occurred during operation due to tire failure

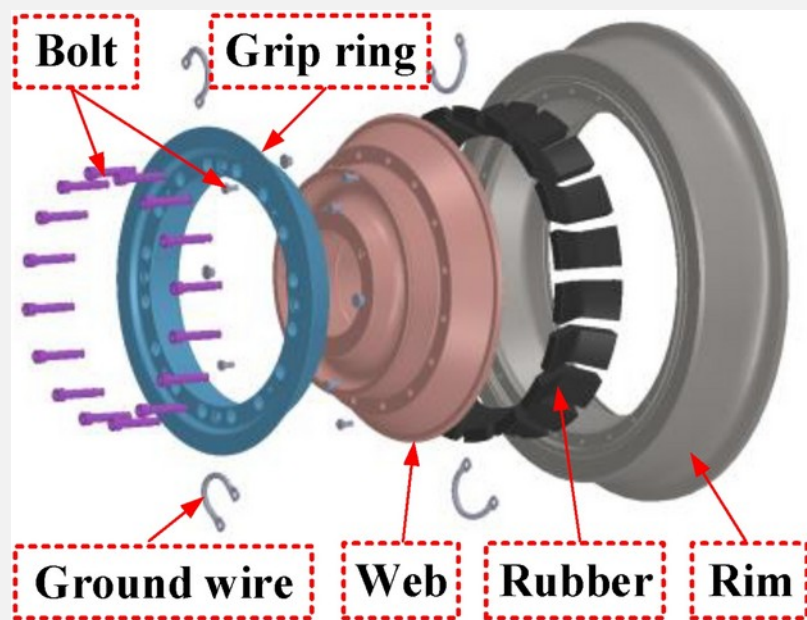
Wheels had not yet reached condemning thickness, but approaching limit

Work Scope:

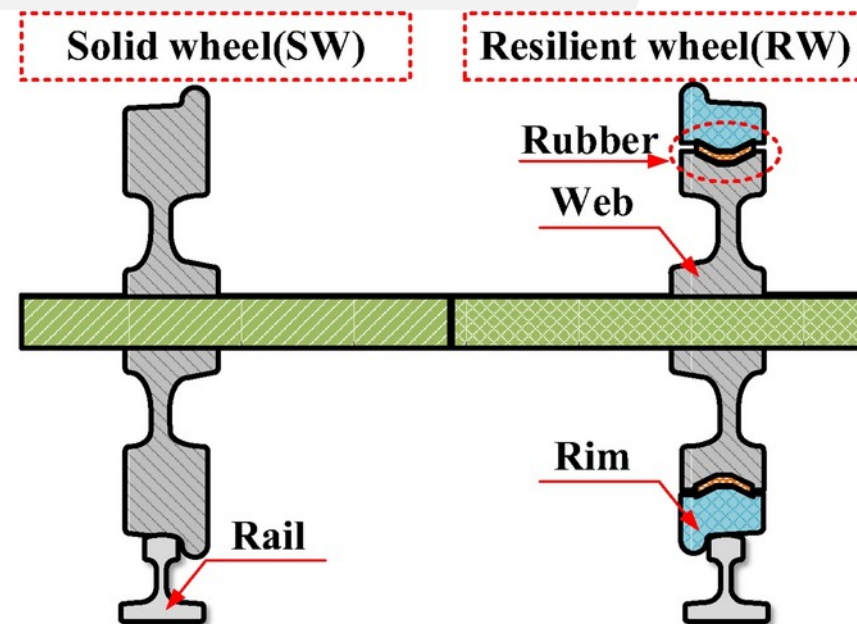
- Fractography of Fracture Surfaces
- Scanning Electron Microscopy (SEM) of Fracture Surfaces
- Chemical Composition Analysis of Tire Material
- Front Face Rim and Cross-Sectional Hardness Measurements



Resilient Wheel Overview



(a)

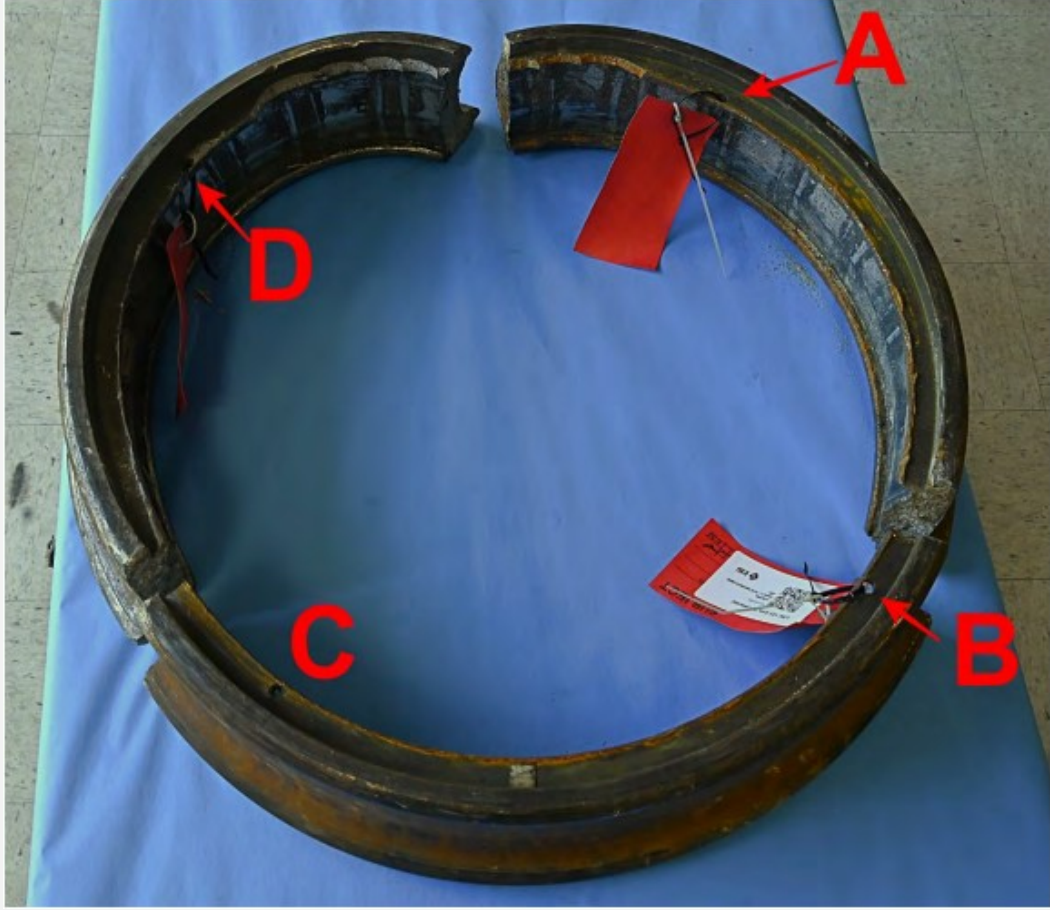


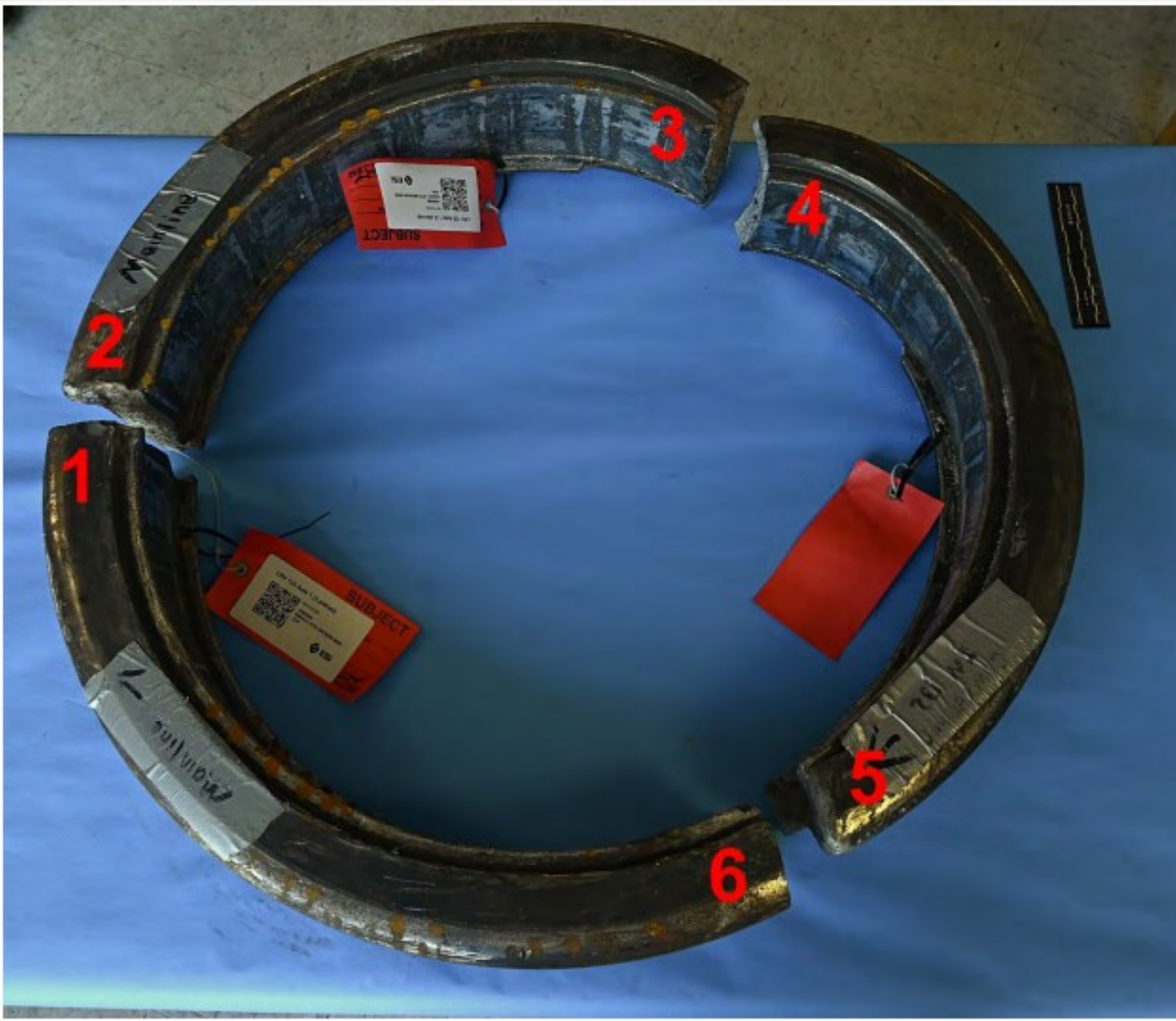
(b)

<https://www.nature.com/articles/s41598-025-86572-0/figures/3>



Fractography of Fracture Surfaces





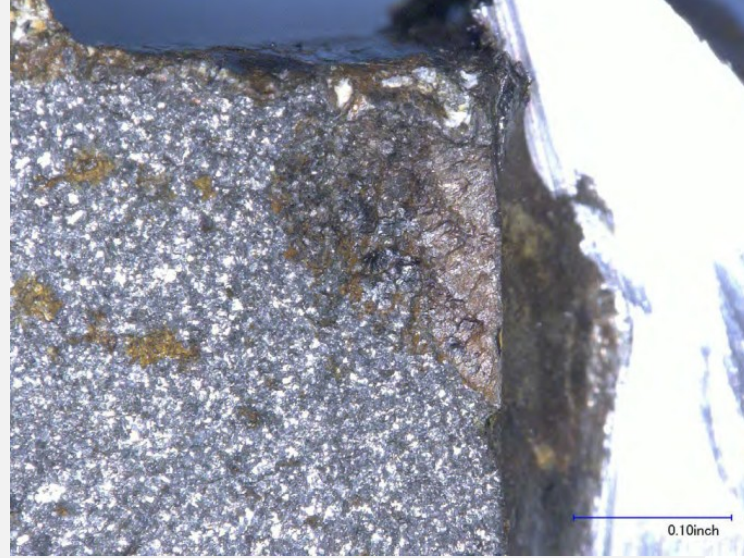
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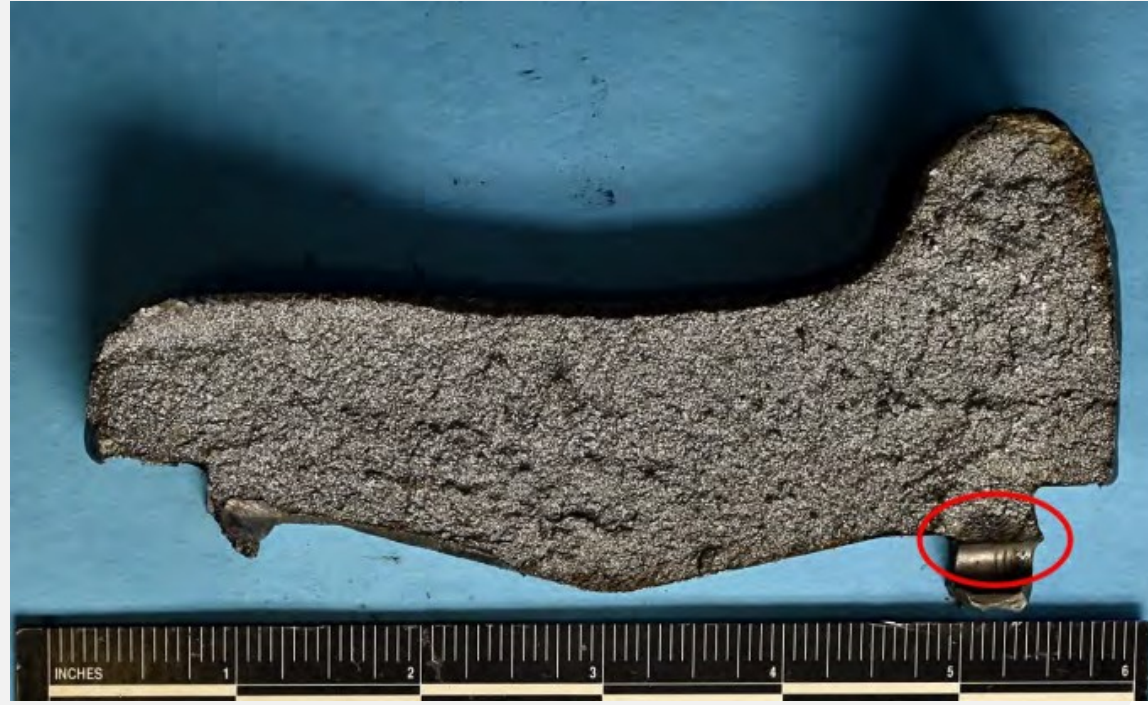
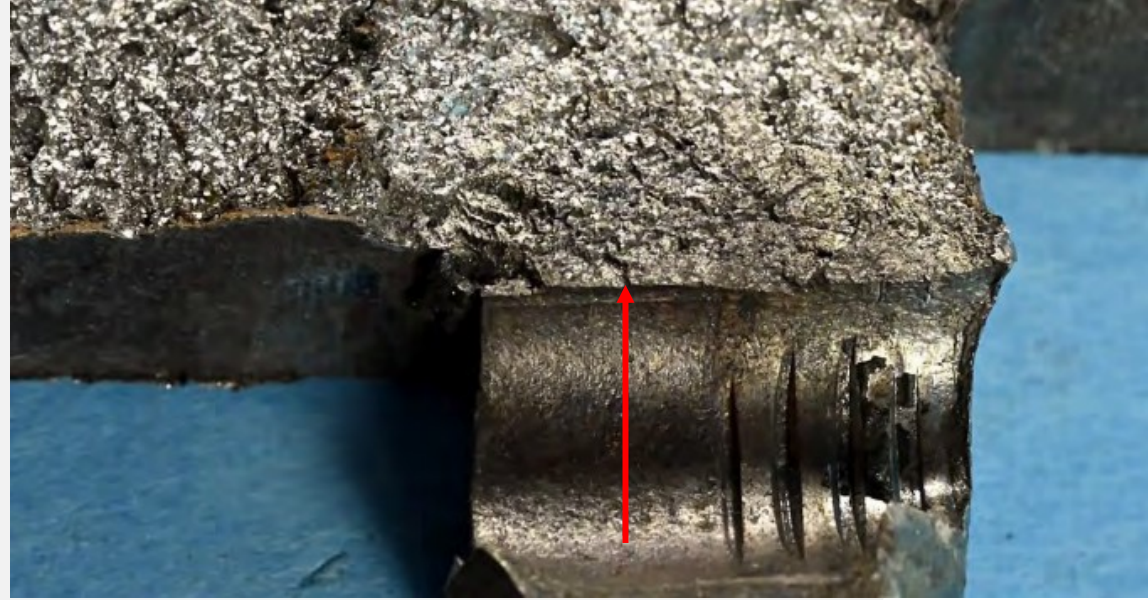




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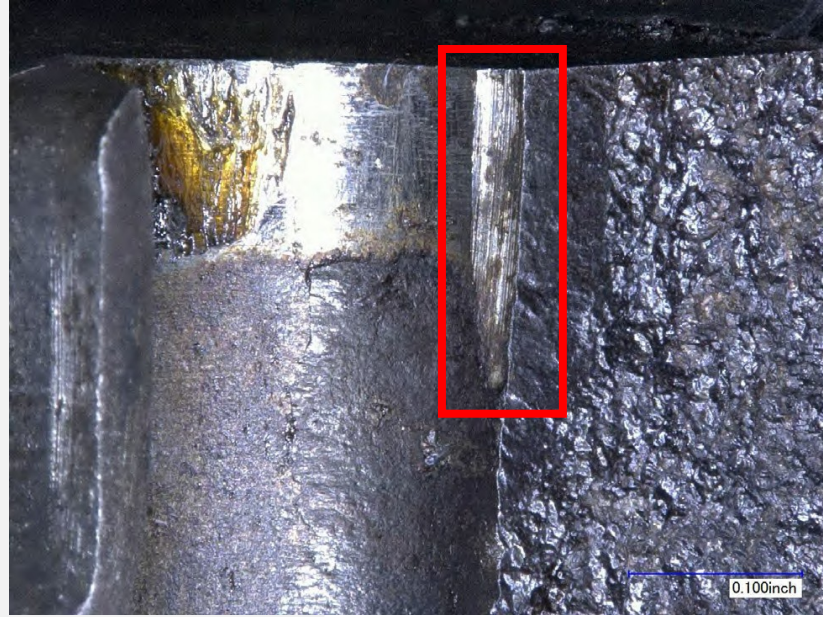
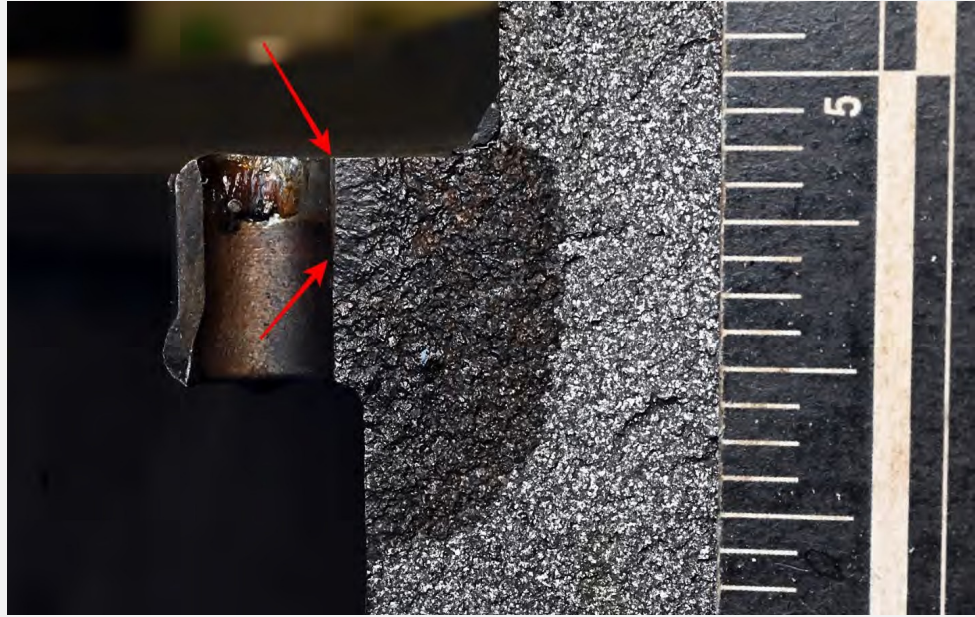
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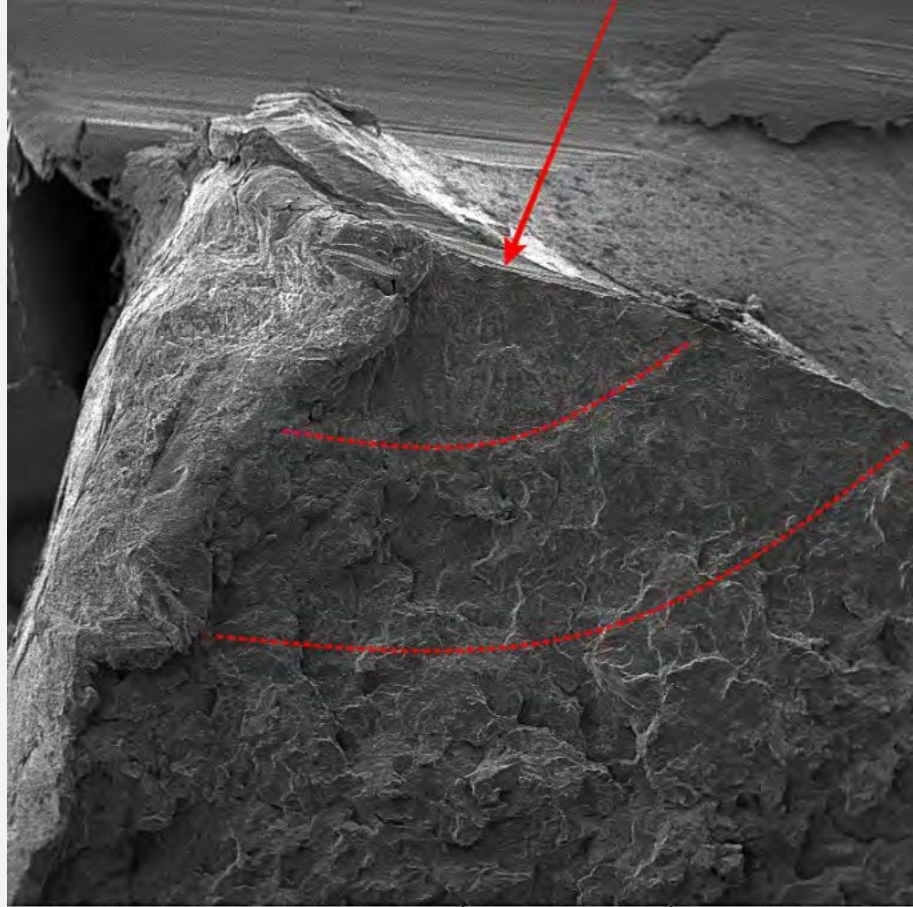
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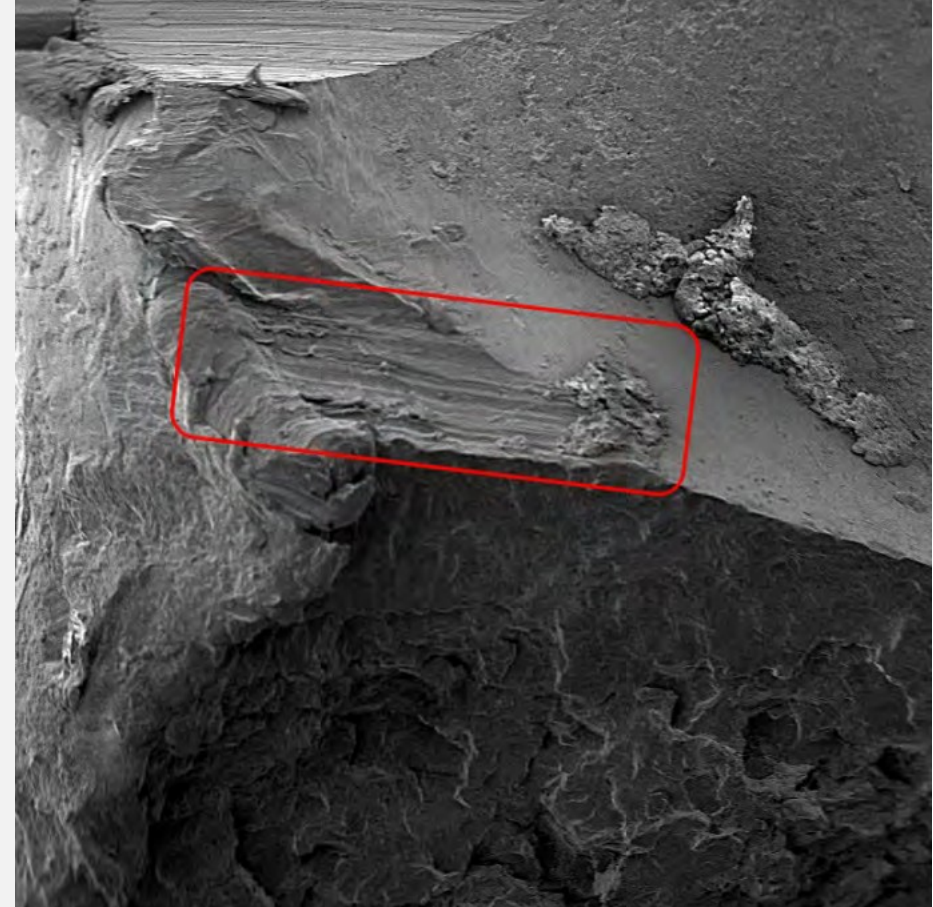




Scanning Electron Microscopy (SEM) of Fracture Surfaces



SEM HV: 10.00 kV WD: 19.34 mm
View field: 4.36 mm Det: SE Detector
SEM MAG: 35 x Date(m/d/y): 07/24/24
VEGA\\ TESCAN
ESI



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VEGA\\ TESCAN
ESI





Additional Testing

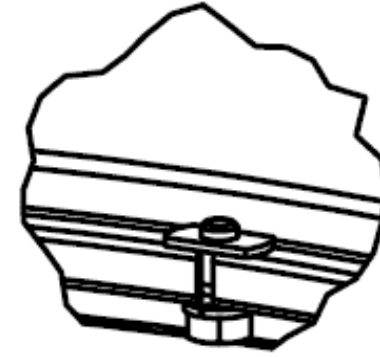


- Metallography
 - Pearlitic Microstructure – No detrimental Phases
- Chemical Composition Analysis
 - Within APTA and Manufacturer Specifications
- Hardness Testing
 - Front Rim Face – Brinell
 - Cross Section Hardness Profile - Rockwell
 - Within APTA and Manufacturer Specifications

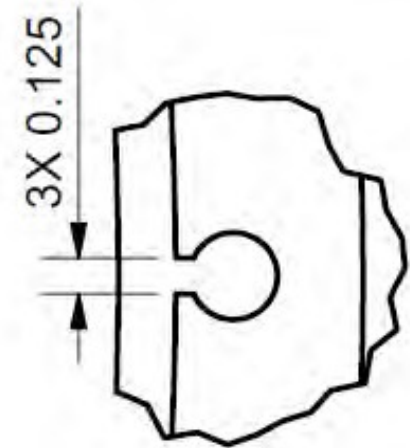


Case Study 1: Conclusions

1. The flange side shunt screw hole initiation location
2. Tool marks were present from cutting of the slot
3. A stress riser from the tool marks resulted in fatigue cracking
4. The fatigue crack size was minor compared to wheel cross section



DETAIL E

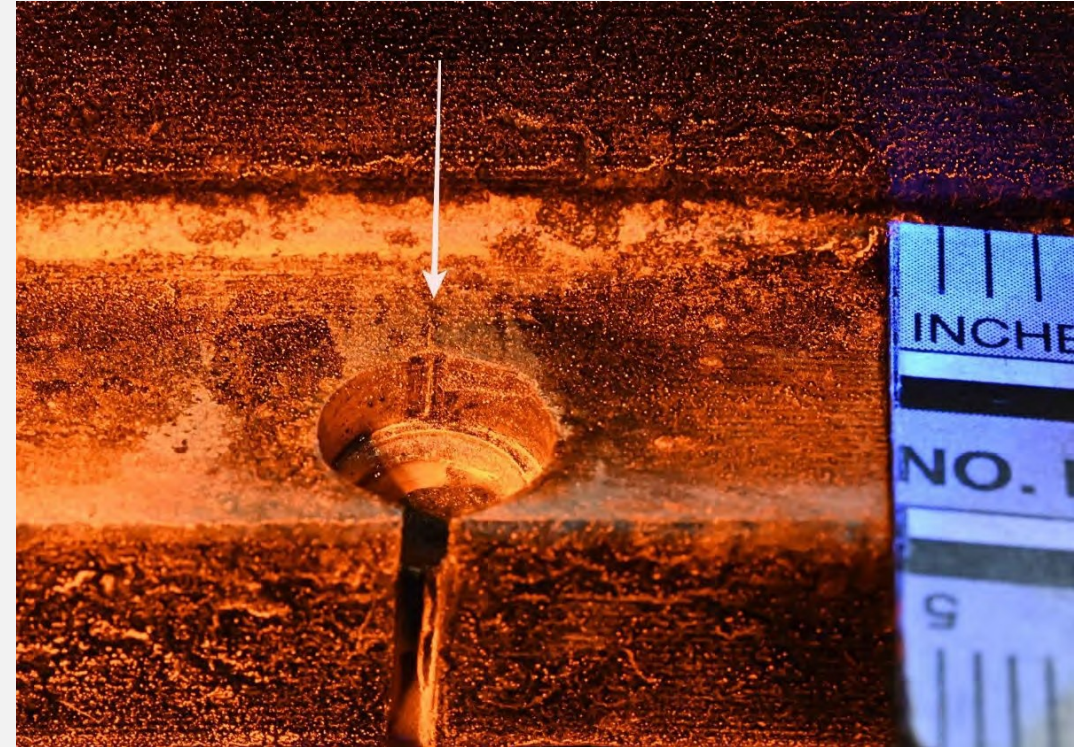


DETAIL OF SLOT
(FLANGE SIDE ONLY)



Case Study 1: Further Work / Prevention

- Inspection of fleet for cracking
- Magnetic Particle testing
- Inspect incoming stock wheels for machine marks





Case Study 2: Failure Analysis of Commuter Wheel



Background:

A wheel was found cracked when returned to the mounting facility. An inspection was performed to determine cause and rule out systemic issue

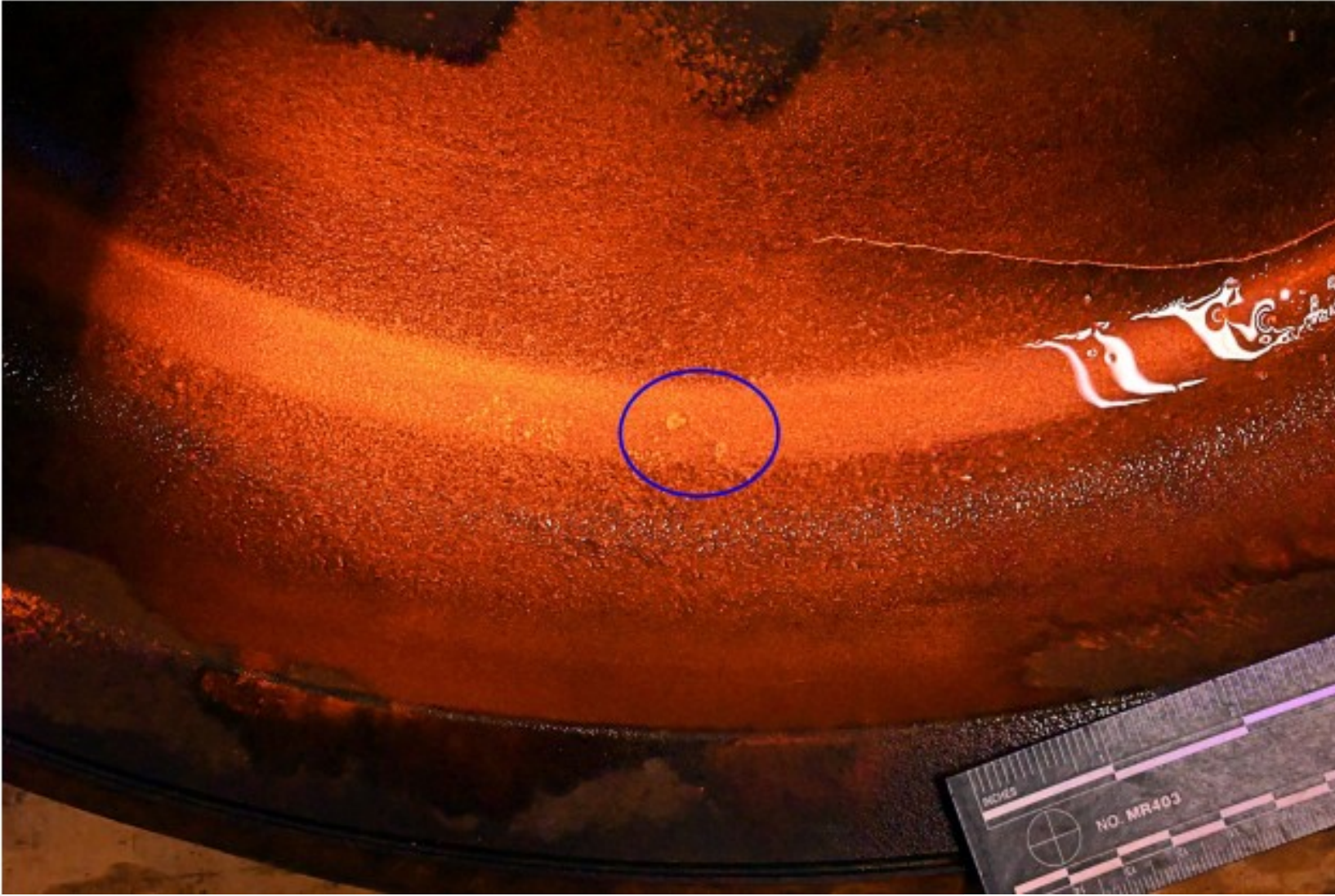
Work Scope:

- Ultraviolet Magnetic Particle Inspection
- Fractography of Fracture Surfaces
- Metallography
- Chemical Composition Analysis
- Microhardness Measurements



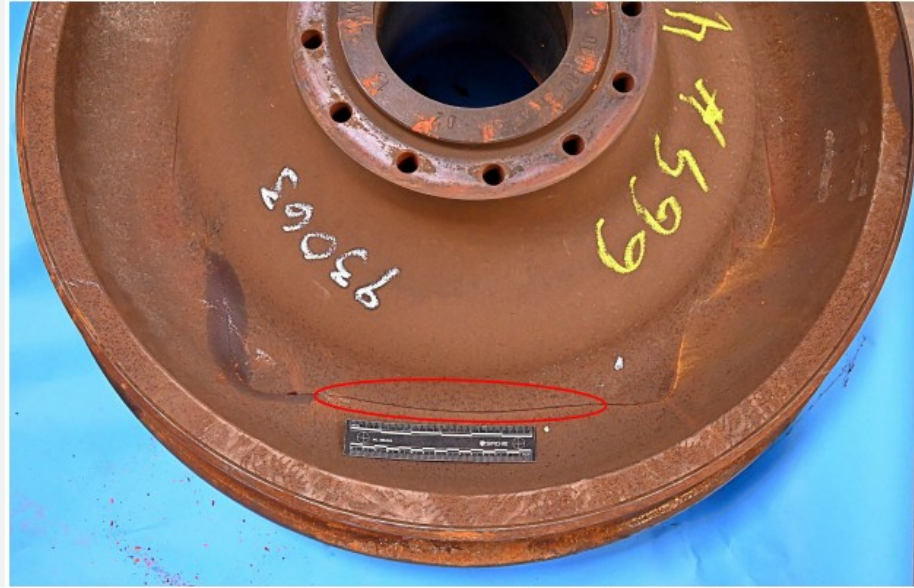


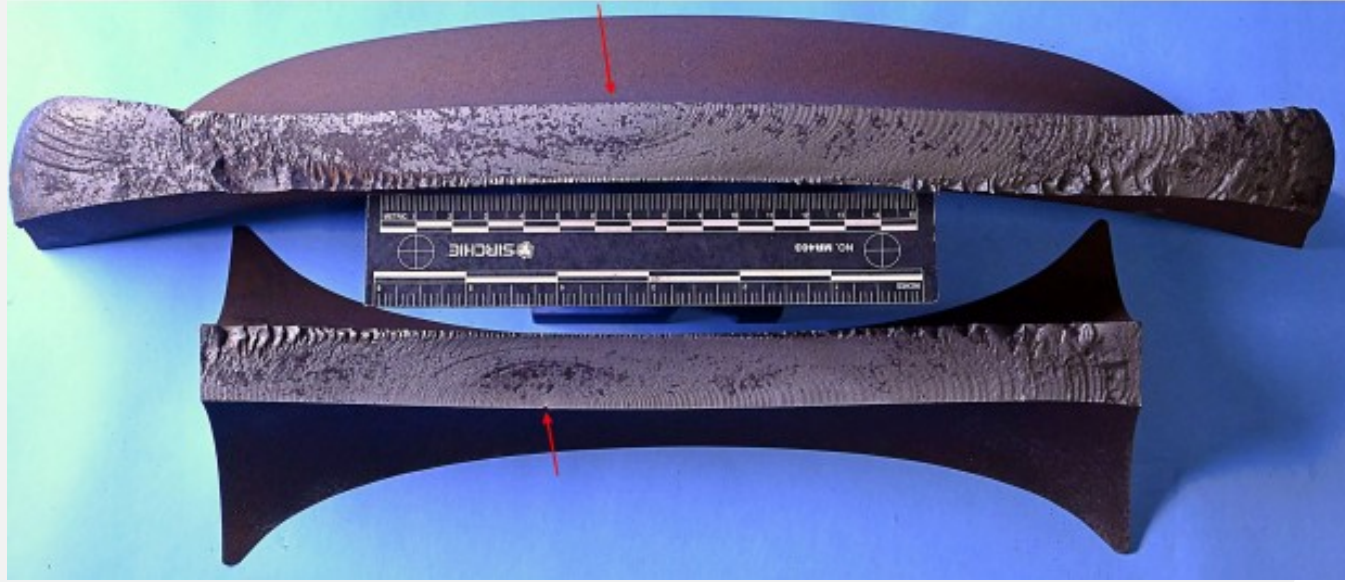
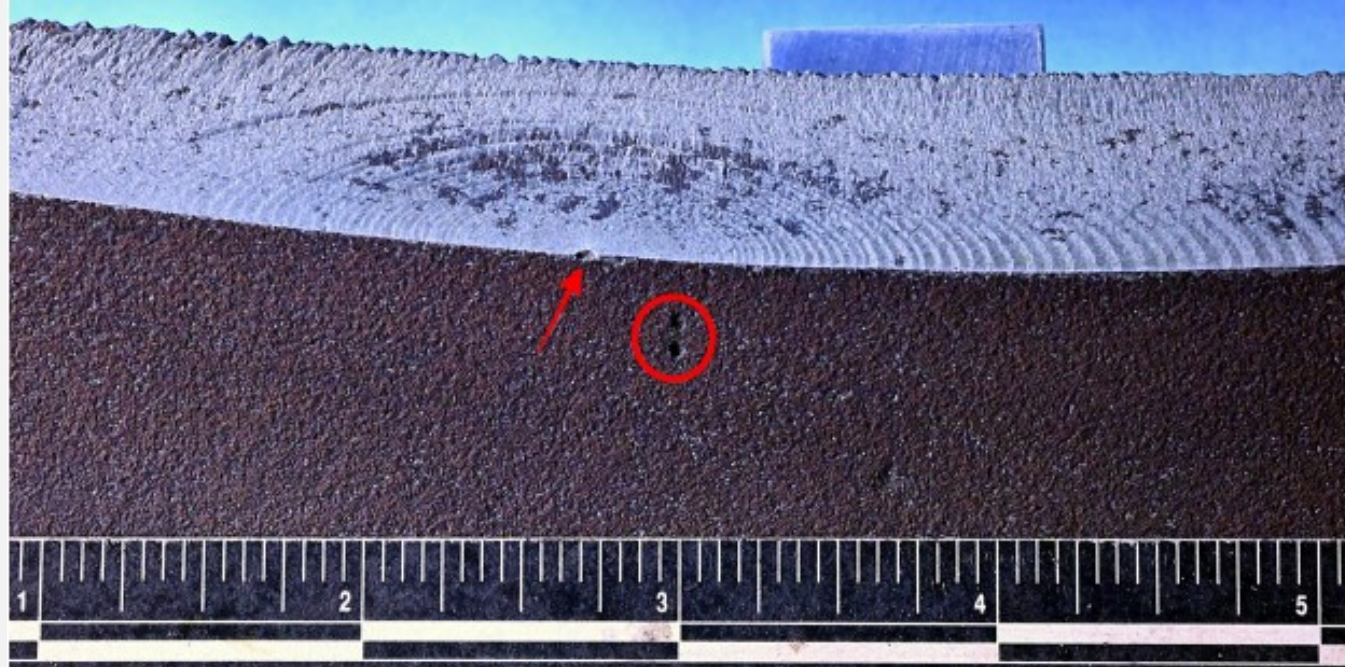
Ultraviolet Mag Particle Inspection





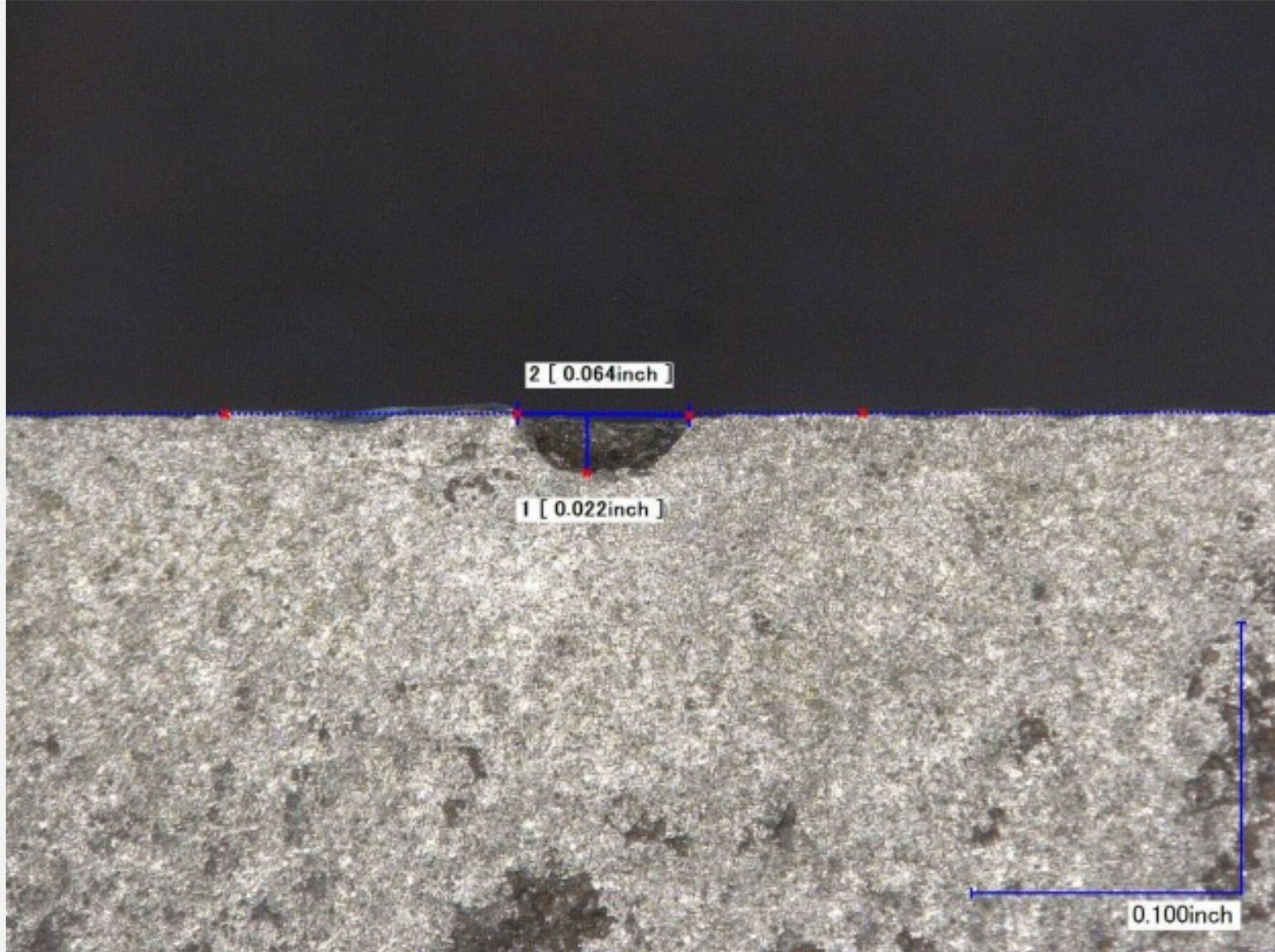
Fractography of Fracture Surfaces





B O S T O







Metallography



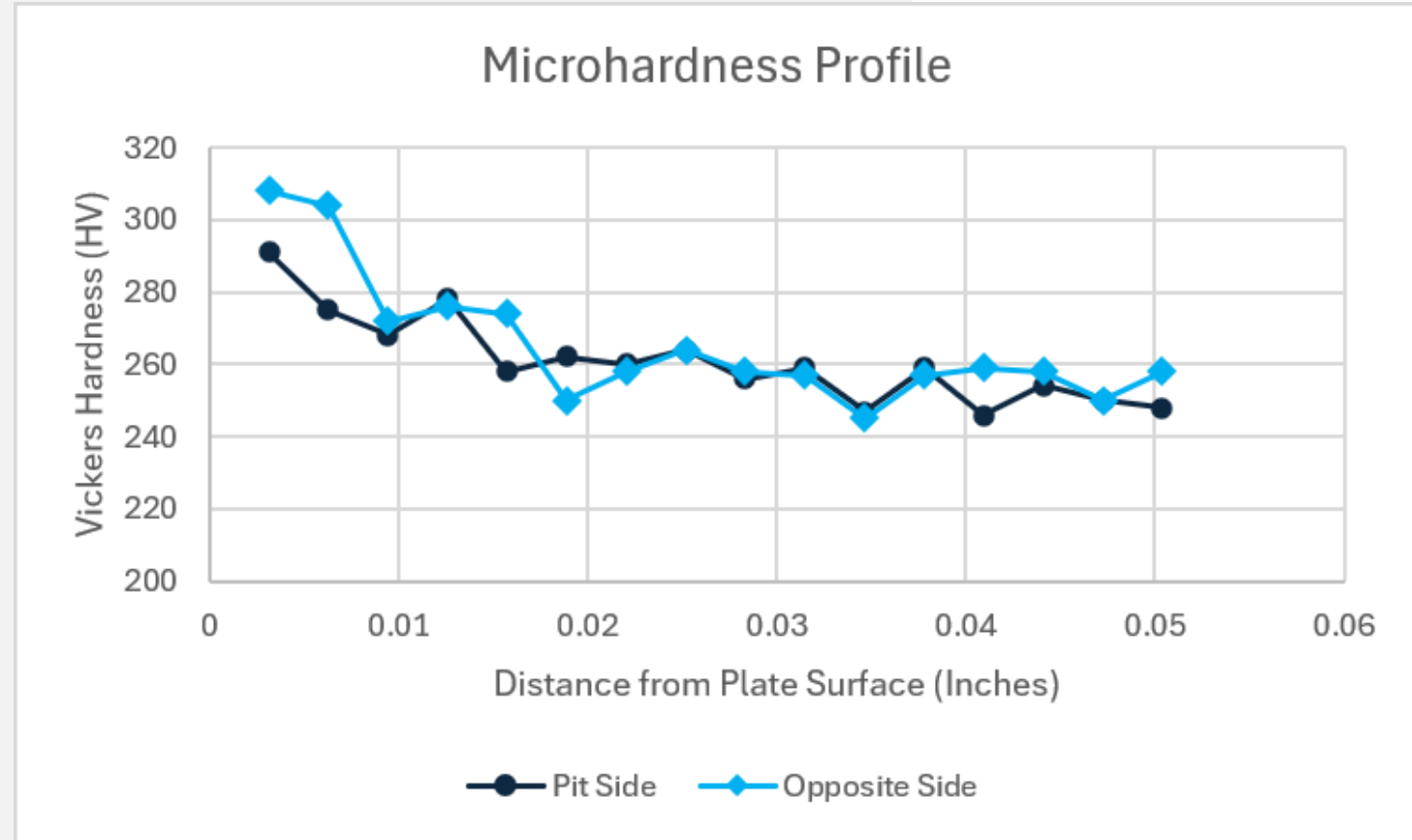






Additional Testing

- Chemical Analysis
 - Within Specification
- Brinell Hardness Testing
 - Within Specification
- Microhardness Testing
 - Confirmed Shot peening had been performed





Case Study 2: Conclusions



1. The wheel cracked due to fatigue cracking that initiated on the front plate of the wheel.
2. The initiation location of the fatigue crack was a corrosion pit that was 0.022 inches deep and 0.064 inches wide
3. The pitting was not widespread



Case Study 3: Failure Analysis of Current Collector



Collector Shoe Overview

Current collector springs provide a controlled, consistent contact force between the collector and the energized rail.

The collector assembly is designed to move relative to the vehicle, while the spring provides the force needed to maintain contact with the rail as the vehicle moves.

If the spring fails, the collector may lose contact with the energized rail, potentially resulting in arcing across the gap.

Arcing can cause localized heating, pitting or burning of the collector and conductor, accelerated wear, and electrical interruptions



Case Study 3: Failure Analysis of Current Collector

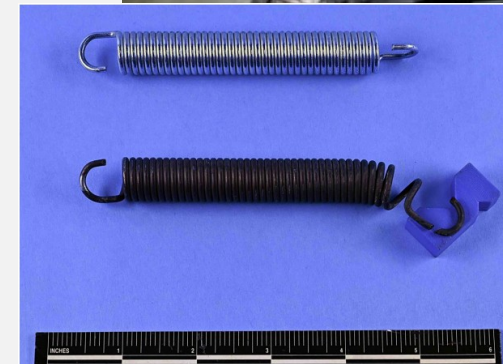


Background:

The failed spring was utilized as part of a current collector device of a rail vehicle.

Work Scope:

- Fractography of Fracture Surfaces
- Scanning Electron Microscopy with Energy Dispersive Spectroscopy
- Metallography and Microstructure Evaluation
- Chemical Composition Analysis
- Microhardness Measurements





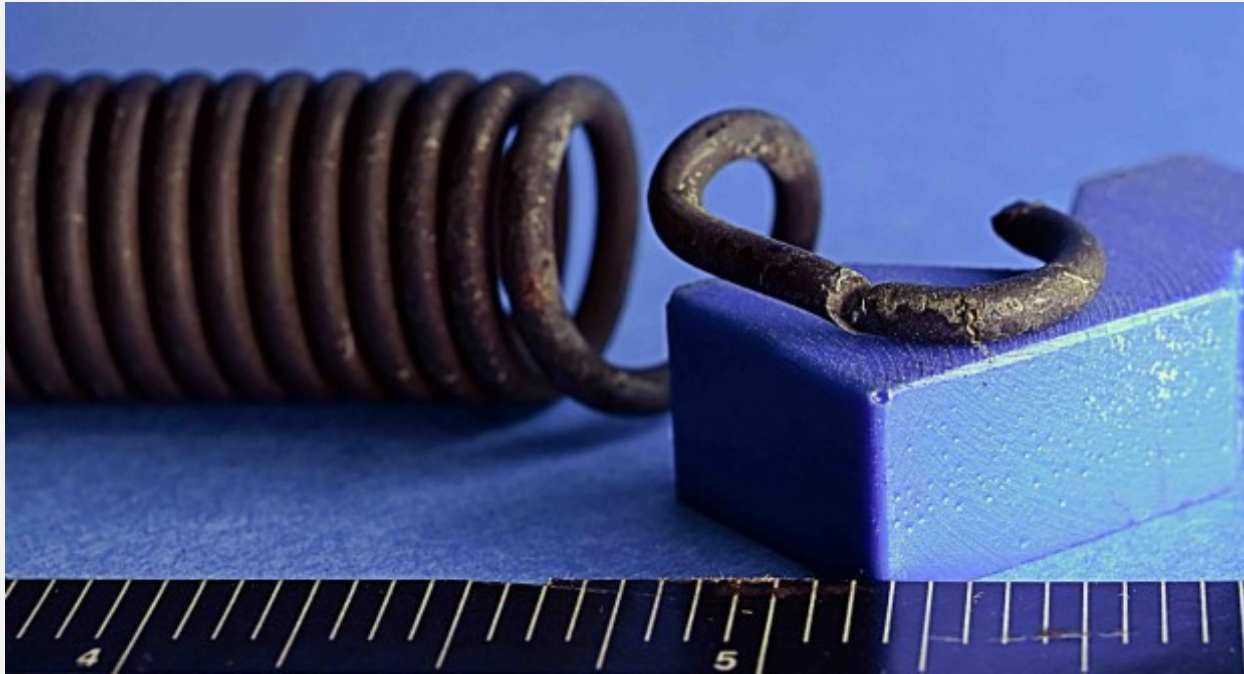
CCD Overview

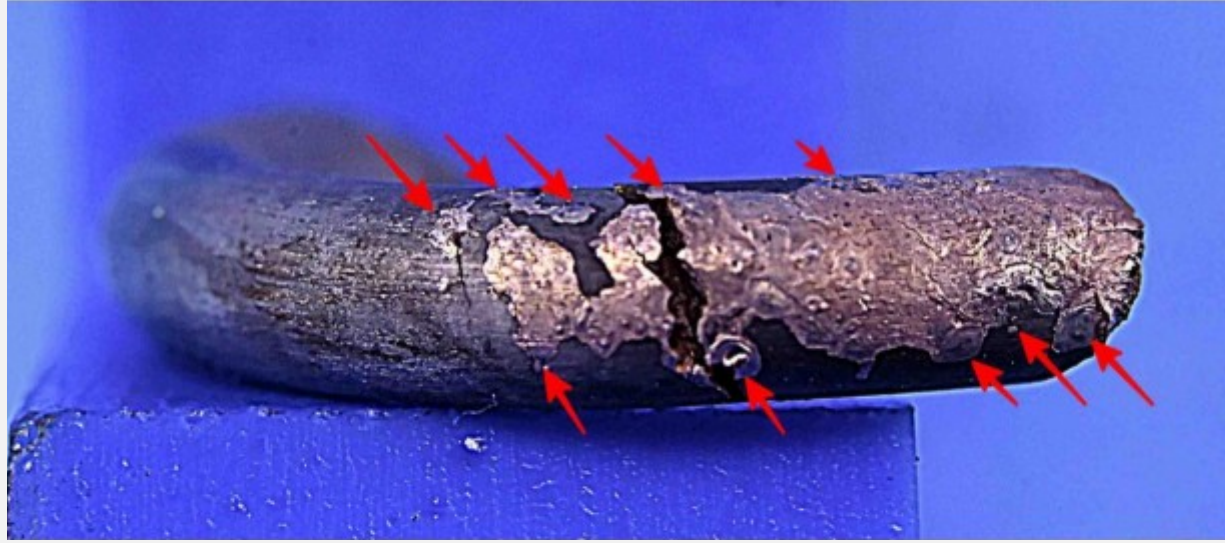
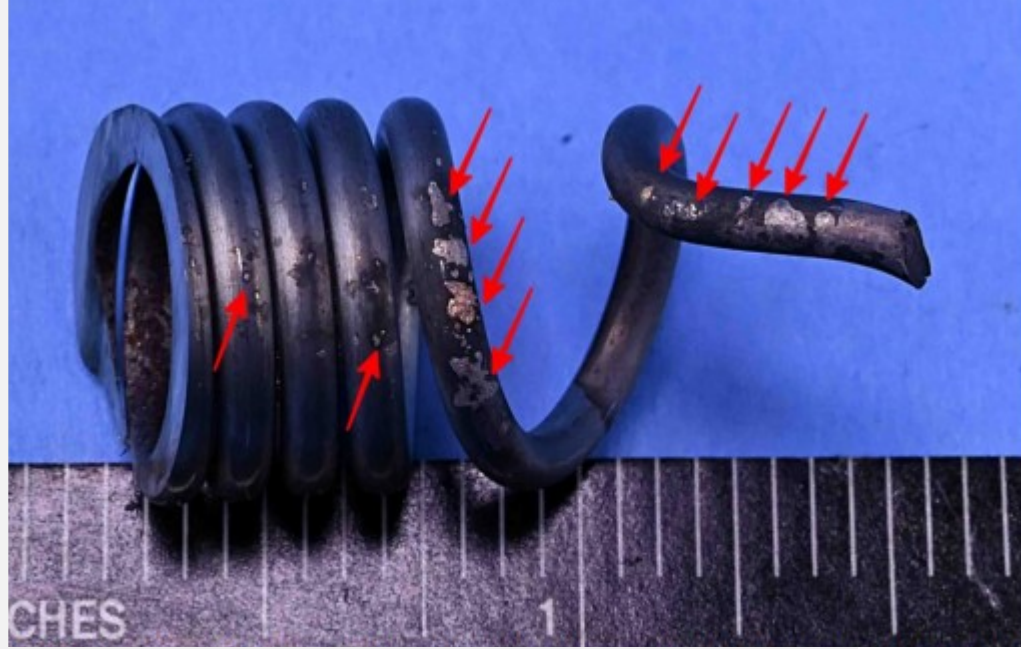


<https://eepower.com/technical-articles/improving-current-collection-with-digital-current-collector-devices/>



Fractography of Fracture Surfaces





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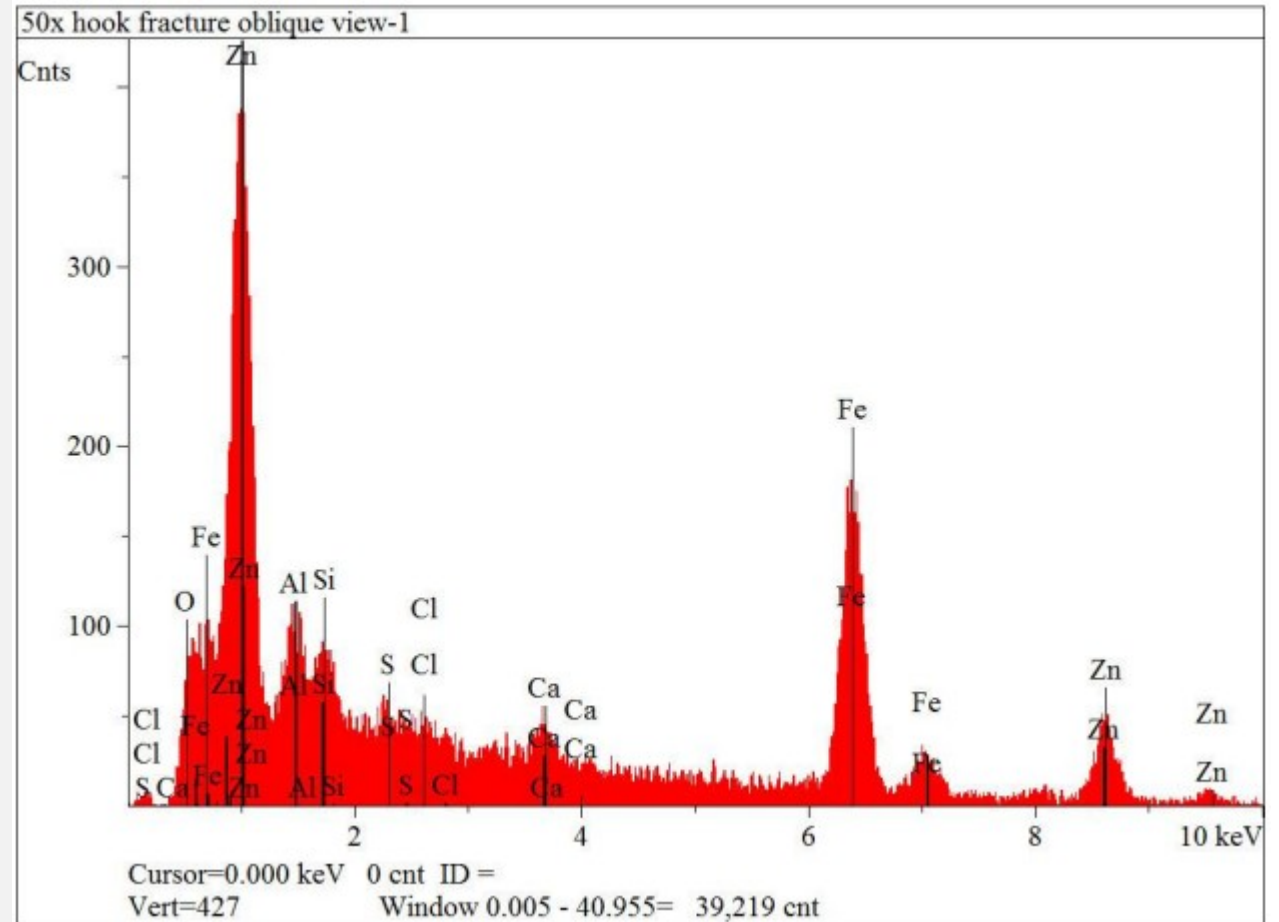
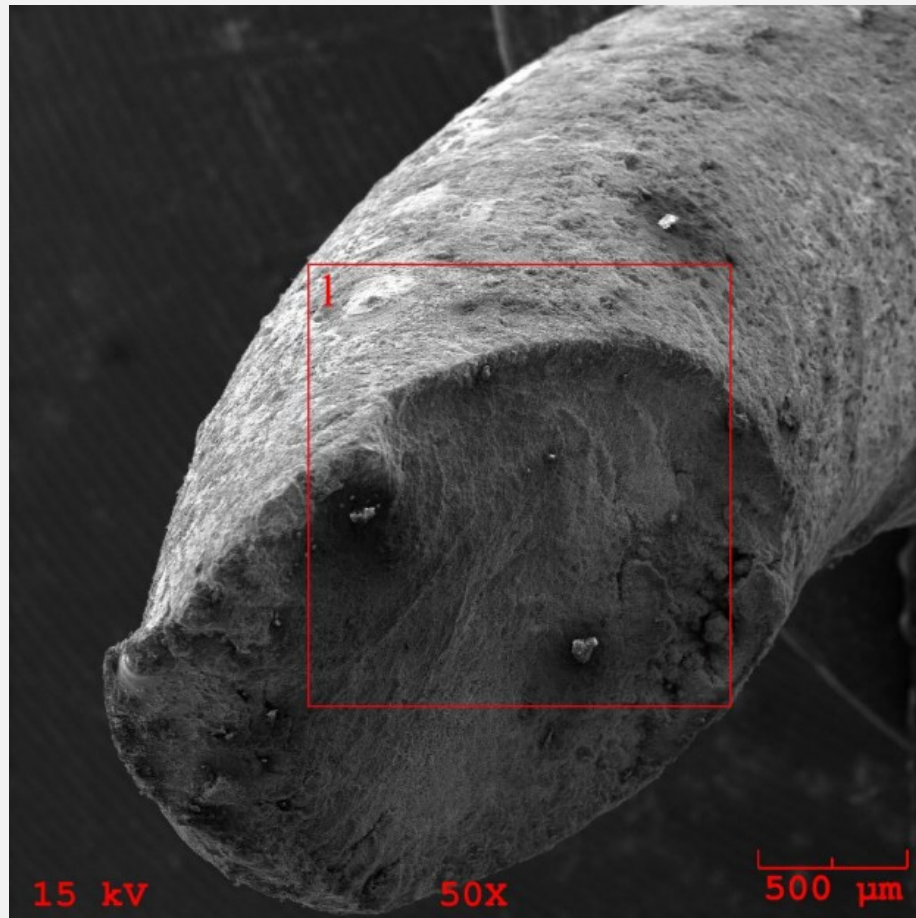


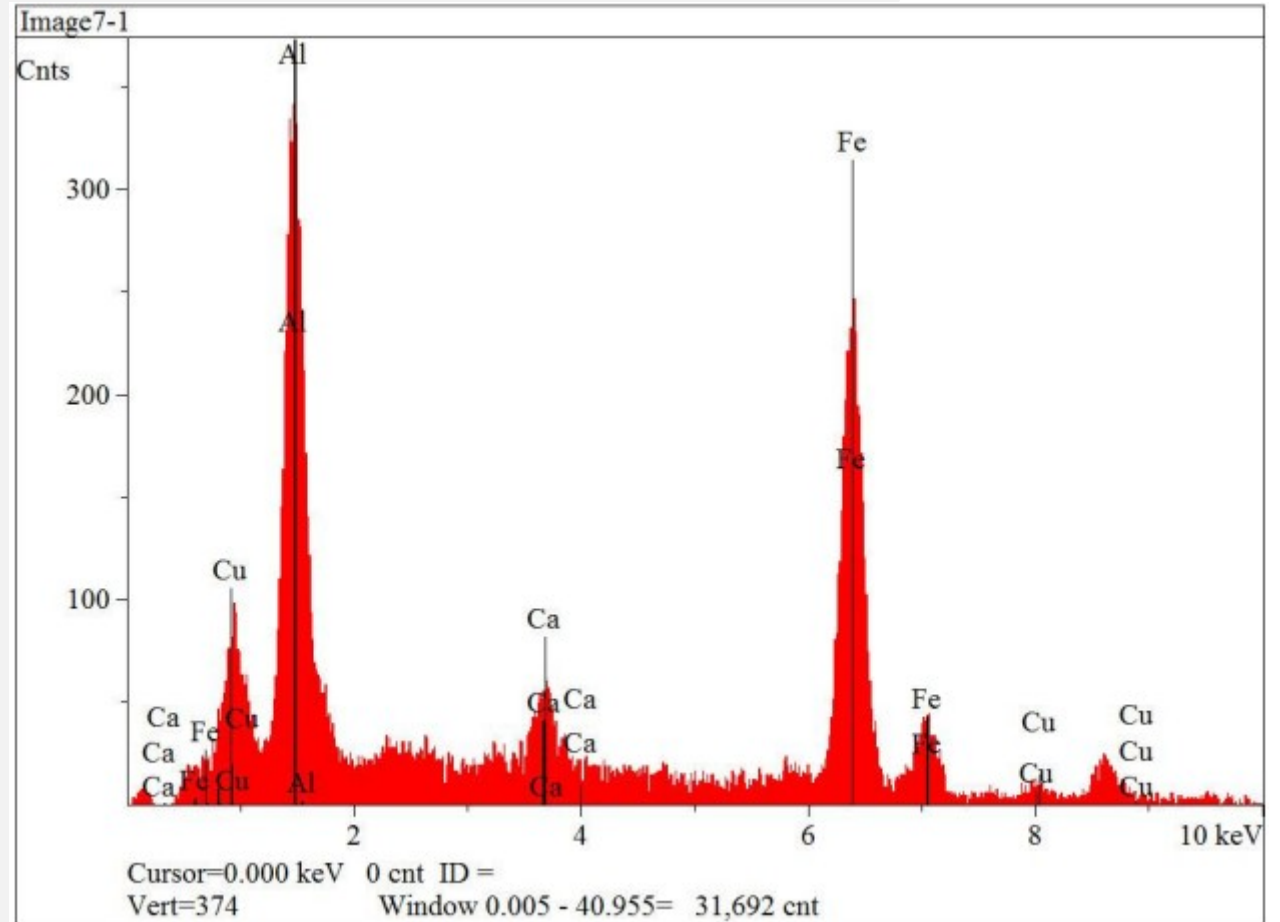
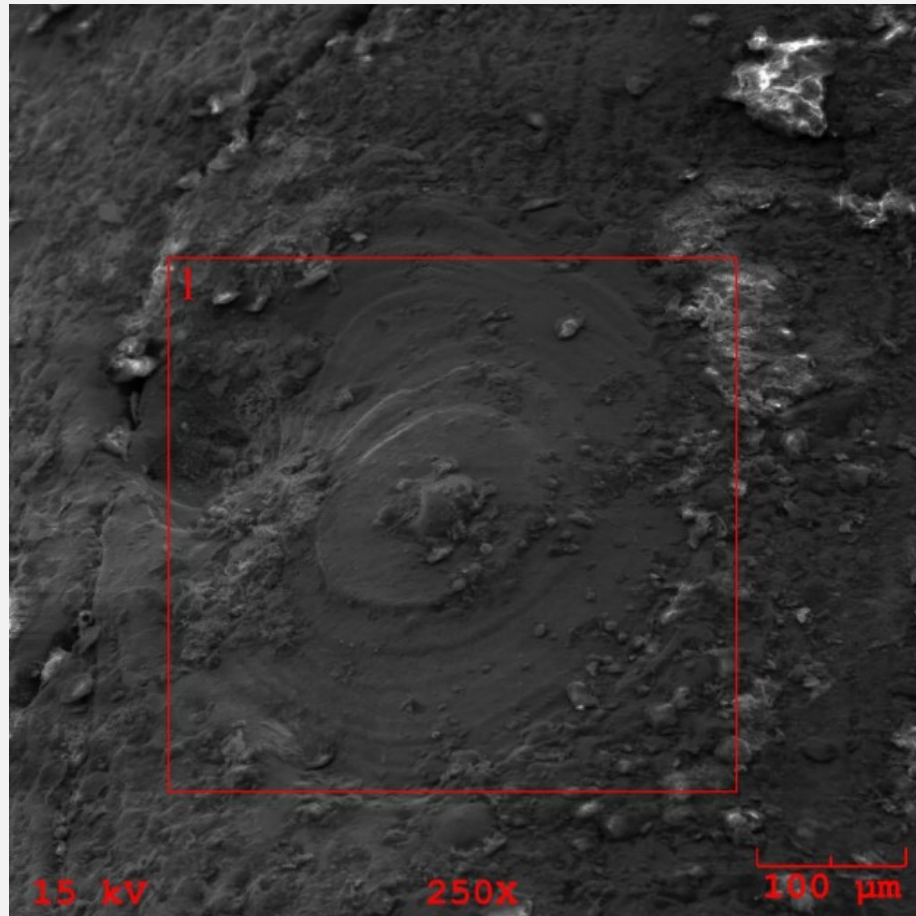


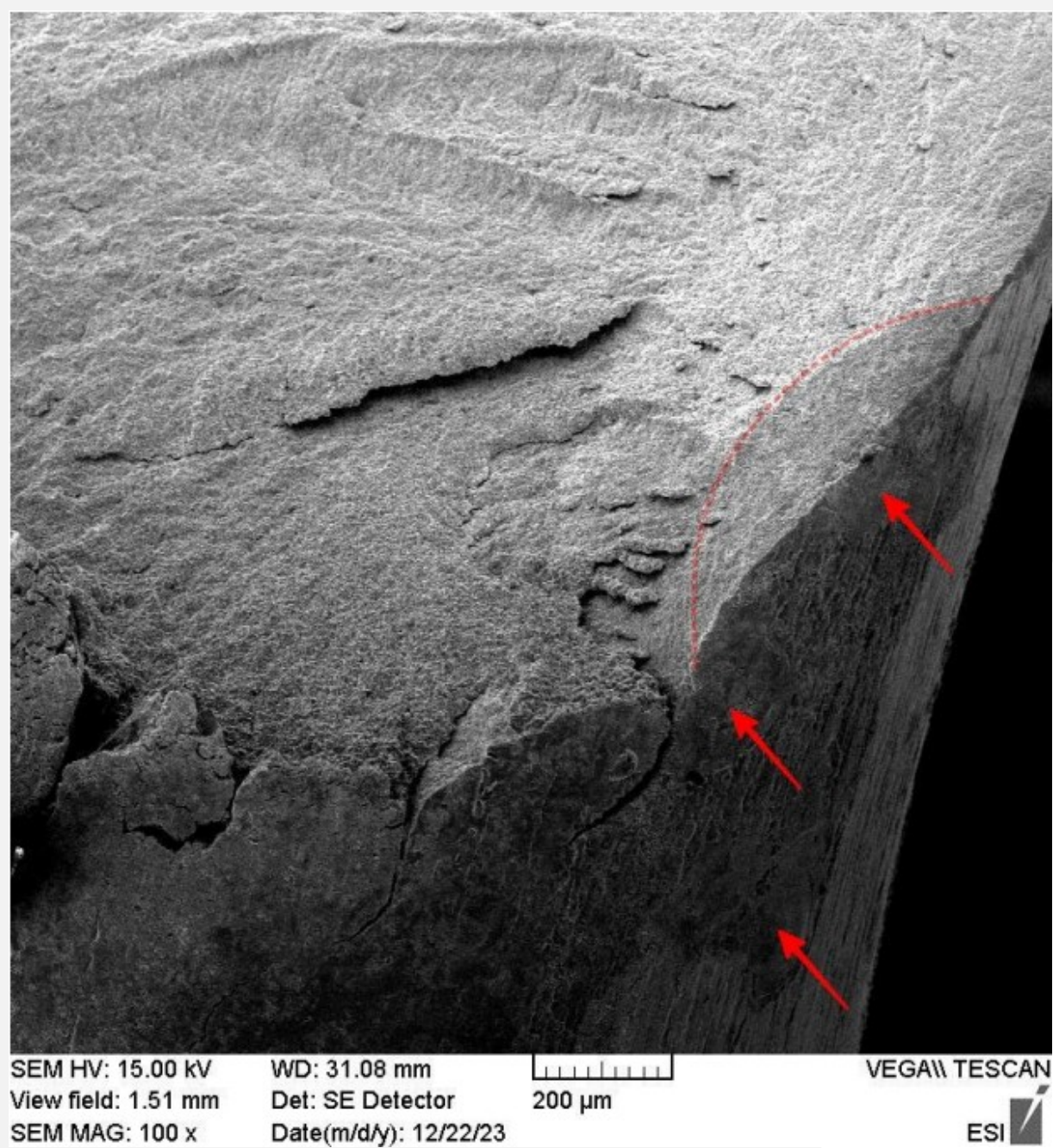




Scanning Electron Microscopy (SEM) and Energy Dispersive Spectroscopy (EDS) of Fracture Surfaces



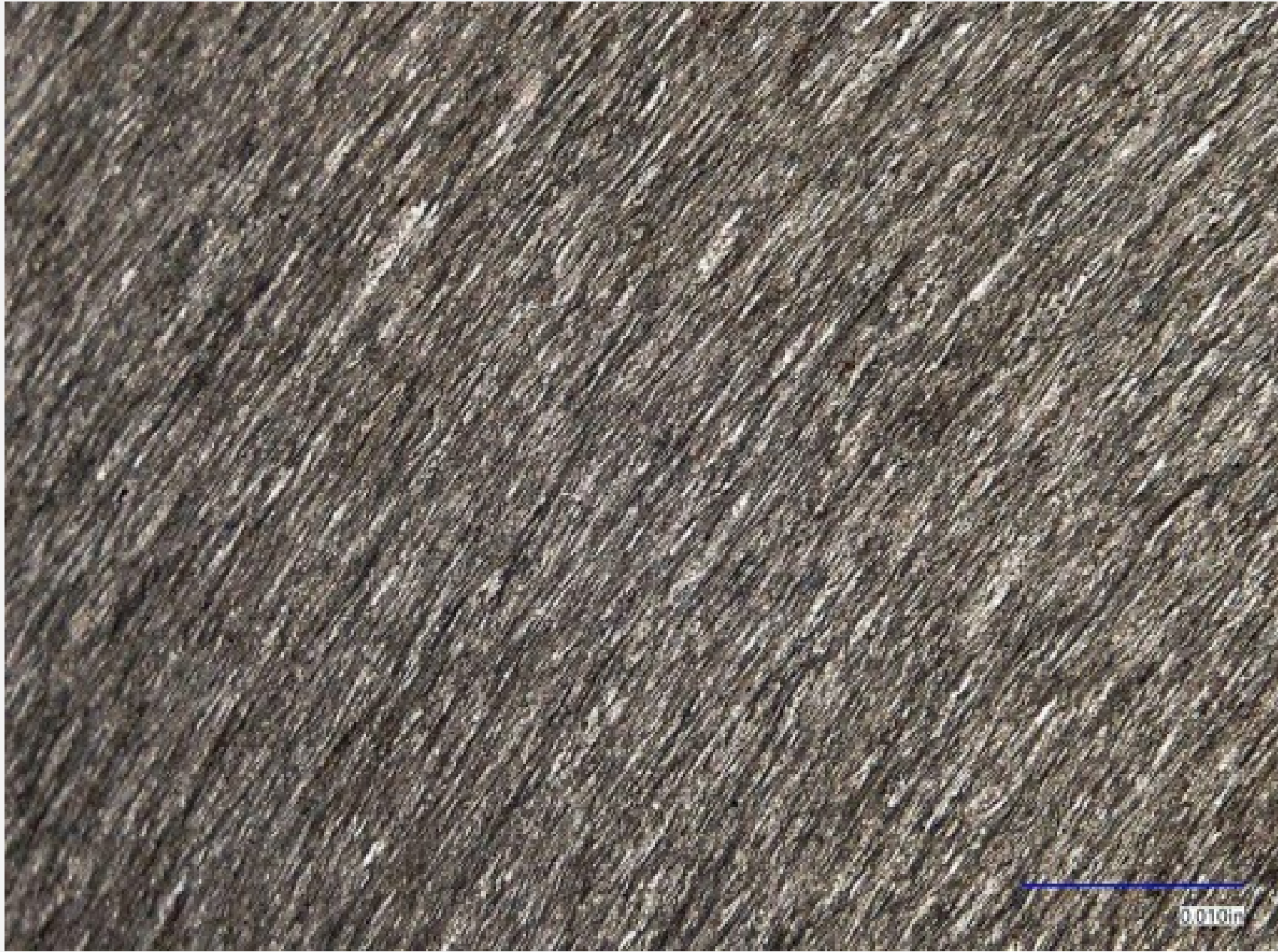


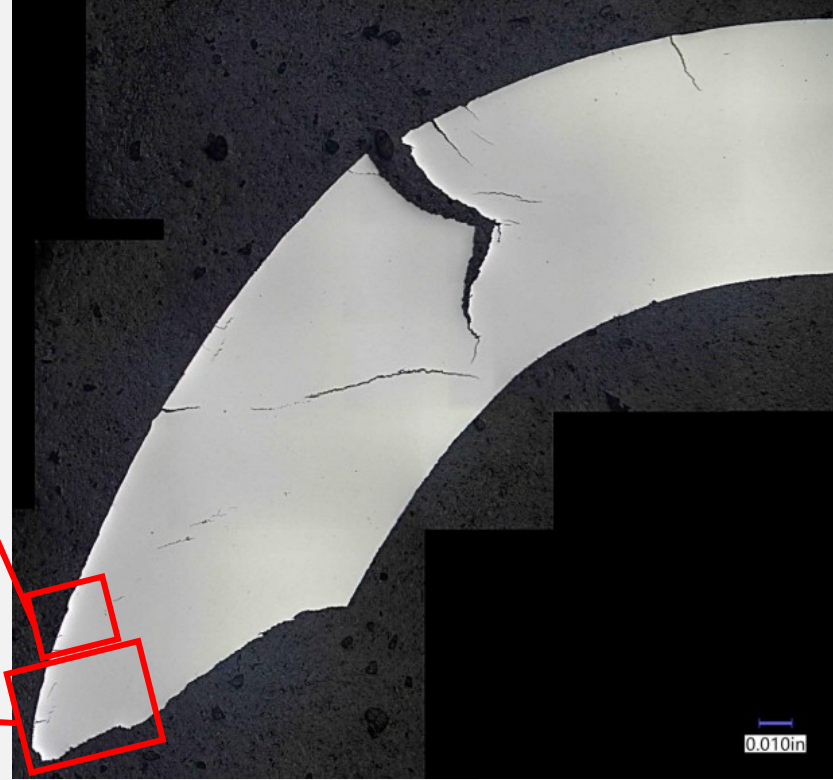
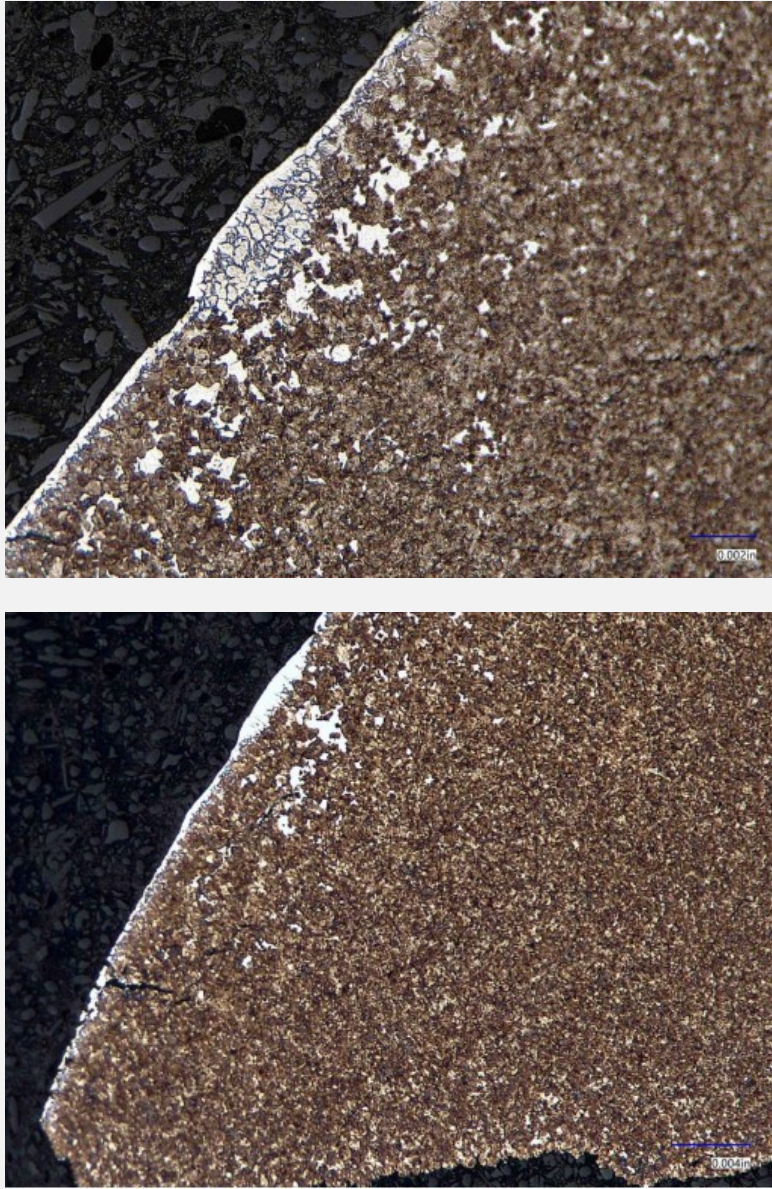




Metallography









Additional Testing

- Chemical Composition Analysis
 - Within Specifications
- Microhardness Testing
 - Varied Results

Location	Hardness Readings	Average Hardness	Approximate Converted Tensile Strength ² (ksi)
Failed Hook Base Metal	532, 546, 546	541	270
Failed Hook Near Inner Surface	309, 309, 319	312	140
Failed Hook Near Arced Outer Surface	431, 448, 474	451	220
New Hook	519, 541, 550	537	270
ASTM A228 Wire Diameter 0.095 inches			Tensile Strength Requirements: 274 - 303



Case Study 3: Conclusions



1. Failure was caused by fatigue cracking initiated from a location of electrical arcing.
2. Un-tempered martensite was present at arc locations
3. Source of arcing not identified, but copper was present at arc locations.
4. The spring ultimately failed due to heat related damage which accumulated from the multiple electrical arcing incidences.
5. The spring had been permanently deformed from its as supplied state.



Conclusions



Thorough component failure investigation can help identify the root cause.

When root causes are identified, corrective and mitigative actions can be taken to help prevent future failures.

Leveraging component failure analysis enables quickly resolving failures to ensure uninterrupted transit service and increased safety.



Thank You!



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